

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
“Jnana Sangama”, Belagavi-590014, KARNATAKA, INDIA



An Internship Report on

Shaale - Vikas

*An internship report submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Computer Science and Engineering** of Visvesvaraya Technological University, Belagavi.*

Submitted by:

AMEEN AHMED(1AM22CS019)

Internship carried out at:

MINDMATRIX

Under the Guidance of:

Internal Guide

Prof. Praveen Kumar B
Assistant Professor,
Dept. of CSE,
AMCEC

External Guide

Tirumal Mutalikdesai,
Program Coordinator,
VTU Internship Program,
MindMatrix.io CL Infotech Pvt. Ltd



Department of Computer Science and Engineering
AMC Engineering College
18th K.M, Bannerghatta Main Road, Bangalore-560083
2025-2026

AMC ENGINEERING COLLEGE

18th K.M, Bannerghatta Main Road, Bangalore-560083

2025-2026

Department of Computer Science and Engineering



CERTIFICATE

Certified that the internship report entitled **“SHAAL – VIKAS”** carried out by **AMEEN AHMED (1AM22CS019)**, bonafide student of AMC Engineering College in partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering of the Visvesvaraya Technological University, Belgaum during the year 2025-26. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report. The internship report has been approved as it satisfies the academic requirements in respect of Internship work prescribed for the said degree.

HOD
Dr. V Mareeswari,
Prof and Head,
Dept. of CSE,
AMCEC

Principal
Dr. Yuvaraju B N,
Principal, AMCEC

Internal Guide
Prof. Praveen Kumar B
Assistant Professor,
Dept. of CSE, AMCEC

Internship
Coordinator
Ms. B Vijaya Nirmala,
Assistant Professor
Dept. of CSE,
AMCEC

External Viva

External Name

Signature with Date

- 1.
- 2.

DECLARATION

I, **AMEEN AHMED**, a student of the 8th semester Department of Computer Science and Engineering, AMC Engineering College, declare that my Internship work entitled “Android Development Using GenAI” is a Bonafide work of mine. Certified that this Internship report is an original report of work done by me under the guidance of internal guide **Prof. PRAVEEN KUMAR B** and under the supervision of external guide **TIRUMAL MUTALIKDESAI** submitted as a part of the Internship Course of Undergraduate Program of Visvesvaraya Technological University, Jnana Sangama, Machhe, Belagavi-590018.

Date:

Signature of the student

Sign. Of Internship Supervisor (College level)

ACKNOWLEDGMENTS

I have a great pleasure in expressing my deep sense of gratitude to founder **Chairman Dr.**

K.R. Paramahamsa and **Executive Vice President Mr. Rahul Kalluri** for having provided me with a great infrastructure and well-furnished labs for successful completion of my internship.

I express my special thanks and gratitude to our Academic Advisor **Dr. Nagaraja R** for providing me with all the necessary advice for successful completion of my internship.

I express my sincere thanks and gratitude to our Principal **Dr. Yuvaraju B N** for providing me with all the necessary support successful completion of my internship.

I would like to extend my special thanks to **Dr. V Mareeswari** Professor and HOD, Department of CSE, for her support and encouragement and suggestions given to me in the course of the internship.

I would like to extend my special thanks to Internship coordinator **Ms. B Vijaya Nirmala**, Assistant Professor, Department of CSE, for her support and encouragement and suggestions given to us in the course of my Internship.

I would like to thank my internship guide **Tirumal Mutalikdesai** for helping and guiding me throughout the internship period.

I am grateful to my guide **Prof. Praveen Kumar B**, Assistant Professor, Department of CSE, AMC Engineering College, Bengaluru for his constant motivation and timely help, encouragement and suggestions.

Last but not the least, I wish to thank all the teaching and non-teaching staffs of department of Computer Science and Engineering, for their support, patience and endurance shown during the preparation of this internship report.

AMEEN AHMED (1AM22CS019)

OFFER LETTER



<https://mindmatrix.io>
+91 80 4974 8471
contact@clinf.com

Internship Offer Letter

To,
AMEEN AHMED

Date : 8 December 2025

Subject: Internship Offer Letter – Android App Development using Gen AI

Dear AMEEN AHMED,

Thank you for exploring internship opportunities with **MindMatrix.io**, a product of **CL Infotech Pvt. Ltd.**, through the **VTU Internship Portal**. We are pleased to offer you an internship with our organization.

This internship will run for **13 to 15 weeks** as per VTU Guidelines. The Internship will provide you with hands-on exposure through guided tasks, structured modules, and practical projects on the **MindMatrix Platform** (<https://lms.mindmatrix.io>).

During the internship, you will be engaged in product application development, supporting feature implementation, testing, preparing documentation, and cross-functional collaboration. These activities will require you to apply your technical knowledge and demonstrate your professional skills effectively.

All **internship policies and professional guidelines** will be provided in the **Internship Manual**, which will be shared with you on the day of program commencement.

As part of the process, you are required to:

- Print this offer letter and fill in the **Faculty Coordinator details** below.
- Obtain **HoD/Internship Coordinator signature with seal**.
- Email the scanned copy back to us within **one week** of receiving it.

For onboarding instructions, please refer to the attached **"Onboarding Steps"** document. We look forward to your participation and wish you a productive internship.

Warm regards,
Tirumal Mutalikdesai
Program Coordinator, MindMatrix.io
contact@mindmatrix.io | 96115 46444

☒ Authorization from Internship In-charge Faculty Or HoD

Name : Dr. V. MAREESWARTI Designation: HOD Dept. of CSE
Contact No.: 9845 198389 Email: hodcse@amceducation.in Dept: CSE

Sign and Seal of HoD: V. MAREESWARTI
17/12/25

#651, Kiran Arcade, 4th Floor, 13th Cross, 27th Main, HSR Layout, Sector 1, Bangalore - 560102

CIN: U72200KA1998PTC023936 Email ID: contact@clinf.com Ph.: +91 96115 46444

ABSTRACT

Shaale-Vikas is an Android-based application developed to bridge the gap between rural schools and their alumni community. Many schools face infrastructure challenges such as damaged classrooms, inadequate furniture, and lack of essential facilities. Due to delays in government funding and lack of awareness among alumni, these issues often remain unresolved or worsen over time.

The proposed system provides a transparent and user-friendly platform where school authorities can post their requirements, along with cost estimates and supporting details. Alumni can view these needs in real time and contribute by pledging support. The application includes features such as a needs dashboard, progress tracking of funds, and a simulated pledge system to demonstrate user participation.

The current implementation focuses on the core functionality, including displaying school needs, tracking progress, and enabling user interaction. Future enhancements include real-time database integration using Firebase, image uploads for better transparency, and secure payment integration.

Shaale-Vikas aims to promote community involvement, improve infrastructure development, and ensure transparency in resource utilization. By leveraging technology, the system provides an effective solution to enhance the quality of education in rural areas through collective support.

CONTENTS

CERTIFICATE	i
DECLARATION	ii
ACKNOWLEDGEMENTS	iii
OFFER LETTER	iv
ABSTRACT	v
1 COMPANY PROFILE	1
1.1 Introduction	1
1.2 Mission	1
1.3 Vision	1
1.4 Services	2
1.4.1 Executive Summary	2
1.4.2 Software Development And Application Design.	2
1.4.3 Enterprise Cloud Solutions And Ai Orchestration.	2
1.4.4 Training and internship Programs	2
1.5 Software Technologies	3
2 ABOUT COMPANY DEPARTMENT	4
2.1 Overview of Department	4
2.2 Organizational Structure	4
2.3 Department Workflow And Methodology.	5
3 TASKS PERFORMED	6
3.1 Work Schedule.	6
3.2 Introduction	8
3.3 System Requirements	8
3.3.1 Functional Requirements	8
3.3.2 Non-Functional Requirements	9
3.3.3 Hardware and Software Requirements	10
3.4. System Design	11
3.4.1 Requirement analysis and agile methodology.	11
3.4.2 UI/UX Flow And Navigation Design	11
3.4.3 Data Structuring And State Management	12

3.4.4	Generative Ai – Assisted Development Workflow	17
3.5	Implementation.....	18
3.5.1	Objectives	18
3.5.2	System architecture implementation	18
3.5.3	Module- wise implementation	19
3.5.4	Transaction management logic	19
3.5.5	Analytics and Reporting.....	19
3.6	Platform	
3.6.1	Programming Language.....	19
3.6.2	Library and Tools	19
4	RESULTS	22
4.1	Project Result.....	22
4.2	Needs Dashboard Functionality	22
4.3	Login Screen Functionality	23
4.4	Home Screen Functionality.....	23
4.5	Donor Hall Of Fame Functionality	23
4.6	Performance Evaluation.....	24
4.7	Advantages Of The System	25
4.8	Limitations	25
4.9	Future Enhancements.....	25
4.10	Conclusion	26
5	COURSE OUTCOMES AND CO-PO MAPPING	
5.1	Course Outcomes	29
5.2	Program Outcomes.....	30
5.3	CO-PO Mapping	31
	BIBLIOGRAPHY	28

LIST OF FIGURES

Fig. No.	Figure Name	Page No.
3.1	Navigation Design	12
3.2	Generative AI – Assisted Development Workflow	13
3.3	System Architecture Implementation	14
3.4	Login Module	15
3.5	Home Screen Module	15
3.6	Pledge Screen Module	15
3.7	Donor Hall of Fame	16
3.8	Firebase Repository Module	17
3.9	Models Module	17
4.1	Needs Dashboard	20
4.2	Login Screen	21
4.3	Home Screen.....	22
4.4	Donor Hall of Fame	22

LIST OF TABLES

Table No.	Table Name	Page No.
1.1	Core Technologies Used At MindMatrix	3
2.1	Department Organizational Structure	4
3.1	14 – Week Internship Schedule	6
3.2	Hardware and Software Requirements	10

CHAPTER 1

COMPANY PROFILE

1.1 Introduction

MindMatrix is an innovation-oriented organization focused on practical software development, experiential learning, and the effective integration of Generative Artificial Intelligence into modern application development workflows. The organization was established with the objective of reducing the gap between academic learning and real-world industry requirements by providing students with hands-on exposure to current technologies and development practices. The company creates an environment that closely resembles real software development ecosystems, allowing interns to understand and participate in the complete development lifecycle. It also emphasizes collaborative learning, enabling interns to work in teams and adopt industry-standard communication and development practices.

1.2 Mission

The mission of MindMatrix is to nurture skilled and industry-ready software developers by providing training in modern tools, technologies, and development methodologies. The organization focuses on transforming academic knowledge into practical skills through project-based learning and continuous technical guidance. A key aspect of this mission is to expose students to real-world problem-solving scenarios where they can design, develop, and implement scalable applications. This ensures that technology is used not only for innovation but also for improving the quality of life in different communities. Additionally, the organization aims to foster critical thinking and adaptability among learners to keep pace with rapidly evolving technological trends.

1.3 Vision

The vision of MindMatrix is to build a strong network of developers and innovators who are capable of creating impactful and scalable software solutions. MindMatrix envisions a future where Generative AI and cloud-based systems become an integral part of the software development process, enabling faster and more efficient application development. From rural healthcare solutions to enterprise-level applications, MindMatrix aims to empower developers to create technology that is both innovative and socially impactful. It also strives to become a leading platform for innovation-driven learning and technological advancement.

1.4 Services

1.4.1 Executive Summary

MindMatrix provides a wide range of services that cover the complete software development lifecycle, starting from idea formulation to final deployment. The organization focuses on combining modern mobile application development practices with advanced technologies such as Generative AI to create efficient and scalable solutions. In addition to technical services, MindMatrix also offers structured professional development programs that help students align their skills with current industry standards and hiring expectations. This integrated approach ensures that both technical and professional competencies are developed simultaneously.

1.4.2 Software Development and Application Design

The organization is actively involved in designing and developing customized software applications based on specific user requirements. This includes building strong backend systems using local data storage solutions such as Room Database, along with creating responsive and user-friendly interfaces using modern UI frameworks. A major focus is placed on developing applications that follow a state-driven approach, ensuring smooth user interactions and efficient performance.

1.4.3 Enterprise Cloud Solutions & AI Orchestration

MindMatrix incorporates cloud technologies and Generative AI into its development workflow to improve efficiency and scalability. By utilizing platforms such as AI-based coding assistants and cloud services, the organization enhances the speed and accuracy of software development processes. Generative AI tools are used to assist in tasks such as code generation, debugging, and automation of repetitive processes. At the same time, strict precautions are taken to ensure data security and reliability. Measures are implemented to prevent issues such as incorrect outputs or security risks, ensuring that all solutions remain robust and dependable.

1.4.4 Training and Internship Programs

A key component of MindMatrix is its well-structured internship and training programs designed to provide practical exposure to students. These programs are divided into multiple stages, beginning with guided learning, followed by independent application development, and concluding with a final project evaluation. Throughout the program, interns are introduced to modern development tools, version control systems, and collaborative workflows used in the industry. Regular mentoring sessions and feedback

reviews help students continuously improve their technical and problem-solving skills.

1.5 Software Technologies

MindMatrix adopts a wide range of modern technologies to support both software development and training activities. These technologies enable the creation of efficient, scalable, and user-friendly applications while also ensuring that students gain exposure to industry-relevant tools. The primary technologies used during the internship are categorized below. These tools and frameworks are carefully selected to align with current industry standards and emerging technological trends.

Table 1.1 Core Technologies Used at MindMatrix

Category	Technologies / Tools
Programming Languages	Kotlin (Primary for Android), Java, Python
Mobile Frameworks	Android SDK, Jetpack Compose, Jetpack Navigation, Material Design 3
Databases	Room Database (SQLite wrapper for local persistence)
AI & Orchestration	Anthropic Claude, Google Antigravity (Agent-first IDE)
Cloud & Enterprise	Google Cloud Platform (GCP), Salesforce CRM, Agentforce
Development IDEs	Android Studio Hedgehog, Antigravity IDE
Management Tools	GitHub, Jira, Lucidchart (UML), Agile Frameworks

Table 1.1 talks about the technologies collectively provide a strong foundation for developing modern Android applications. They support various aspects of development, including user interface design, data management, cloud integration, and AI-assisted coding. Exposure to these tools during the internship helps students build practical skills that align with current industry requirements.

CHAPTER 2

ABOUT COMPANY DEPARTMENT

2.1 Overview of Department

The internship was conducted under the Mobile Application Development and AI Integration Department. This department focuses on developing modern Android applications while integrating Generative AI to improve the overall software development process. It emphasizes the use of current technologies and tools within the Android ecosystem to build efficient and scalable applications. The department follows Agile development practices, promoting iterative progress, continuous feedback, and effective team collaboration instead of traditional development approaches. Interns are encouraged to take ownership of their projects, starting from problem identification to final implementation and deployment. The department also supports continuous learning by encouraging the use of online platforms and enterprise tools alongside project work. A significant focus of the department is on solving real-world problems, particularly those affecting rural and underserved communities. Interns are guided to design applications that can operate in low-connectivity environments, ensuring accessibility and practical usability. This approach helps in developing meaningful solutions that contribute to digital empowerment and social impact. Additionally, the department fosters innovation by encouraging experimentation with new ideas and emerging technologies to enhance application effectiveness.

2.2 Organizational Structure

The department maintains a flat, collaborative organizational structure designed to foster mentorship and rapid skill acquisition. Roles are clearly distributed to emulate a real-world software production team, with explicit responsibilities at each tier:

Table 2.1 Department Organizational Structure

Role	Responsibilities
Internship Coordinators	Define overall objectives, explain the structured phases of the internship, and introduce interns to company practices and developer expectations. They also track weekly progress and conduct evaluation reviews at the end of each phase.

Role	Responsibilities
Technical Mentors	Experienced professionals who lead technical sessions, review code, and assist interns in implementing key concepts such as database handling, UI state management, and safe usage of Generative AI tools.
Interns / Junior Devs	Responsible for developing applications independently, preparing documentation, designing system models, and building capstone projects such as Shaale – Vikas , while also maintaining professional profiles.
QA & UX Reviewers	Provide feedback on usability and design during the development phase. They evaluate aspects such as accessibility, interface clarity, and overall user experience, especially for applications targeting rural users.

Table 2.1 describes the four-level organizational structure of the Mobile Application Development and AI Integration Department at MindMatrix, outlining the responsibilities of coordinators, mentors, interns, and reviewers.

2.3 Department Workflow and Methodology

The department follows a structured Agile-based workflow that includes regular sprint cycles, weekly meetings, and periodic demonstrations. Interns work closely with their assigned mentors and are required to maintain version-controlled code using a shared GitHub repository. Proper coding practices are followed, and all changes undergo review before being integrated into the main project branch. This organized workflow ensures that even individual student projects such as Shaale – Vikas are developed with a professional approach, including proper documentation, version tracking, and systematic development practices. The use of issue tracking and structured updates helps maintain clarity and consistency throughout the project lifecycle. Daily progress updates and stand-up meetings further enhance communication and ensure that challenges are identified and resolved promptly. This workflow also helps interns develop time management and teamwork skills essential for real-world software development environments.

CHAPTER 3

TASKS PERFORMED

3.1 Work Schedule

The internship followed a well-structured 14-week development plan from February 2, 2026 to May 10, 2026. The entire duration was divided into three major phases — Guided Learning, Skill Development, and Capstone Project Implementation — to ensure a gradual transition from basic concepts to independent application development. The schedule was designed to build strong fundamentals before moving towards the final capstone project, Shaale - Vikas . The detailed week-wise activities are presented in Table 3.1.

Table 3.1 14-Week Internship Work Schedule

Timeline	Tasks & Milestones Completed
Week 1 (Feb 2 - 8)	Phase 1: Orientation & Introduction. Overview of MindMatrix environment and internship structure. Understood expectations of a software developer role and improved professional profile setup.
Week 2 (Feb 9 - 15)	Introduction to Cloud & CRM tools. Explored Salesforce Trailhead modules and basic cloud computing concepts. Earned badges through guided learning paths.
Week 3 (Feb 16 - 22)	Android Setup. Installed Android Studio and configured SDK and emulator. Learned project structure including folders such as java, res, and manifest.
Week 4 (Feb 23 – Mar 1)	Kotlin & UI Basics. Practiced Kotlin fundamentals and started Jetpack Compose. Developed basic apps like Business Card, Dice Roller, and Tip Calculator.
Week 5 (Mar 2 - 8)	UI Enhancement & Data Handling. Built BMI Calculator and other apps using data classes, lists, and functions. Applied Material Design principles.
Week 6 (Mar 9 - 15)	Phase 2: Independent Development. Created a Task Management app with CRUD operations using Room Database and basic UI layouts.

Timeline	Tasks & Milestones Completed
Week 7 (Mar 16 - 22)	Agile & Planning. Studied Agile methodology and prepared system design documents including PRD and UML diagrams. Explored Google Cloud basics.
Week 8 (Mar 23 - 29)	AI Tools Introduction. Worked with Generative AI tools and implemented navigation between multiple screens using Compose Navigation.
Week 9 (Mar 30 - Apr 5)	Capstone Selection. Evaluated multiple problem statements and selected Shaale - Vikas. Prepared initial project plan and documentation.
Week 10 (Apr 6 - 12)	Phase 3: Architecture Design. Defined system requirements and designed UI layout. Created initial structure of the application.
Week 11 (Apr 13 - 19)	Core Module Development. Implemented Needs page functionality and basic data handling logic.
Week 12 (Apr 20 - 26)	Feature Expansion. Added login module and implemented estimate amount and progress tracking with sample data.
Week 13 (Apr 27 - May 3)	Project Development Phase. Developed the core modules of the Shaale-Vikas application including the Needs Dashboard, pledge system, and progress tracking functionality. Improved UI design consistency and implemented user interaction features.
Week 14 (May 4 - 10)	Testing & Finalization Phase. Performed testing on emulator and real devices, fixed bugs, and ensured smooth functionality. Prepared project documentation including problem statement, system design, implementation details, and finalized the application for submission and presentation.

Table 3.1 represents the complete 14-week internship schedule, outlining weekly learning activities, development milestones, and final deliverables leading to the successful completion of the Shaale – Vikas capstone project.

3.2 Introduction

The main objective of the tasks performed during the internship was to gain proficiency in modern Android application development by building a series of progressively advanced applications, ultimately leading to a socially impactful capstone project named **Shaale-Vikas**. The tasks were structured to gradually increase in complexity, beginning with basic user interface design and simple logic-based applications to understand the core concepts of Android development. As the internship progressed, the focus shifted towards handling real-time data, integrating location-based services, and implementing cloud storage using Firebase. Further advancement involved developing features such as reporting and tracking school infrastructure needs, image-based data capture, and storing and managing data using Firebase services. Finally, all the acquired knowledge was integrated into the **Shaale-Vikas** project, a multi-module mobile application designed to bridge the gap between rural schools and alumni by enabling transparent reporting and resolution of school needs. The application incorporates features such as a needs dashboard, progress tracking, pledge simulation, and an impact visualization system to encourage community participation. This structured progression ensured a strong foundation in both fundamental and advanced concepts of Android development. It also helped in building confidence to independently design, develop, and deploy complete mobile applications that address real-world problems.

3.3 System Requirements

3.3.1 Functional Requirements

The capstone project **Shaale-Vikas**, along with its supporting modules, required the following functional capabilities:

- **Shaale-Vikas Core System:** Must allow the headmaster (admin) to create and manage school infrastructure needs such as repairs, furniture, and basic facilities. Each need should include a title, description, estimated cost, and optionally an image. The system should track the funding progress and display it in real time using a progress indicator. Alumni users should be able to view listed needs and pledge support (simulated contribution system).
- **Needs Management System:** Must allow users to view all posted school needs in a structured dashboard format. Each entry should display details such as need title, cost, collected amount, and progress percentage. The system should support dynamic updates of data and ensure smooth interaction using list-based

UI components.

- **Pledge and Contribution System:** Must allow alumni users to pledge support for specific needs. Each pledge should update the collected amount and reflect changes in the progress of the selected need. The system should simulate contributions for demonstration purposes.
- **Progress Tracking System:** Must display real-time progress of each infrastructure need using visual indicators such as progress bars and percentage values, helping users easily understand the current funding status. The system should dynamically update contribution progress whenever a pledge is made, ensuring transparency and clear visualization of collected and remaining amounts .
- **Local Data Handling:** Must store application-related data locally using mechanisms such as SharedPreferences and structured data models for efficient access and temporary data management during the prototype stage. This helps maintain smooth application performance and enables quick retrieval of user interactions and contribution updates.
- **Basic Applications (Foundational Learning):** Must support user input, process logic correctly, and display results dynamically to build understanding of Android UI and functionality.

3.3.2 Non-Functional Requirements

- **UI/UX & Responsiveness:** The application should use Jetpack Compose for dynamic UI updates and smooth performance without lag.
- **Design Standards:** Consistent use of Material Design principles for layout, typography, and color schemes.
- **Accessibility:** User-friendly interface with clear labels, sufficient spacing, and readable design for all users.
- **Performance:** Fast response time and efficient data handling.
- **Security (GenAI):** Proper validation of AI-generated outputs and careful integration into the application.
- **Availability:** The system should be available for use at all times, provided there is internet connectivity, ensuring uninterrupted access to reporting features.
- **Maintainability:** The application should be designed in a modular and

structured manner, making it easier to update, debug, and extend with new features.

- **Compatibility:** The application should run on different Android devices with varying screen sizes and API levels (minimum SDK 24 and above).
- **Usability:** The system should be easy to learn and use, allowing users to quickly understand how to report waste and navigate through the application.

3.3.3 Hardware and Software Requirements

Table 3.2 Hardware and Software Requirements

Component	Specification
Hardware	A system with minimum 8 GB RAM (16 GB recommended) and a processor capable of running Android Studio smoothly. Virtualization support should be enabled for running Android Emulator. A physical Android device (API 24 or above) can be used for testing.
Operating System	Windows 10/11, macOS, or Linux
Development IDE	Android Studio (latest version)
Programming Language	Kotlin
UI Design	XML Layouts (Android UI Design)
Database	Firebase Firestore (Cloud NoSQL Database)
APIs Used	Google Maps API, Location Services API
Libraries & Tools	AndroidX Libraries, Material Design Components, Google Play Services
Build System	Gradle

Testing Tools	Android Emulator / Physical Device
---------------	------------------------------------

The software requirements listed above in **Table 3.2** provide the necessary environment for developing and running the application efficiently. Android Studio is used as the primary development platform, along with Java/Kotlin for programming and XML for user interface design. The application uses Android UI components and adapter-based data handling to display and manage school infrastructure needs efficiently. Supporting libraries such as AndroidX and Material Design components are used to ensure a modern, responsive, and user-friendly interface. Local data handling mechanisms are utilized during the prototype stage for smooth functionality and testing. The overall technology stack enables efficient development, testing, and execution of the **Shaale-Vikas** application while also providing flexibility for future enhancements such as Firebase integration and real-time data synchronization.

3.4 System Design

3.4.1 Requirement Analysis and Agile Methodology

Before starting implementation, system planning was carried out using Agile methodology. A Product Requirements Document (PRD) was prepared for **Shaale-Vikas** to define the objectives, features, functional requirements, and overall scope of the application. The application requirements included management of school infrastructure needs, progress tracking, simulated pledge system, and transparent visualization of contributions to ensure accountability and community participation. The system was designed using an iterative development approach, where individual features such as the needs dashboard, pledge functionality, progress tracking, and basic UI components were implemented step by step. Each module was developed and tested independently before integration into the main application. This ensured flexibility in development, continuous improvement, and efficient handling of changing requirements during the implementation phase. The Agile methodology also helped in dividing the project into smaller development cycles, making testing, debugging, and feature enhancement easier and more manageable. Regular reviews and incremental improvements ensured that the application remained stable, user-friendly, and aligned with the intended project objectives throughout the development process.

3.4.2 UI/UX Flow and Navigation Design

The front-end of the application was developed using XML-based layouts in Android Studio. The interface was designed using modern UI components such as LinearLayout, ScrollView, ListView, and Card-based views to ensure smooth rendering and user-friendly interaction. Navigation was implemented using multiple activities, allowing users to move between the Dashboard Screen and the Add Need Screen. The Dashboard displays all school needs along with cost and progress details, while the Add Need screen allows entry of new requirements. A consistent and simple design pattern was maintained across all screens to improve usability and clarity. User interactions such as button clicks and list selections were handled using event listeners, with feedback provided through Toast messages. The design also considered usability aspects such as clear text labels, proper spacing, and intuitive button placement to ensure ease of use for all users, including non-technical stakeholders.

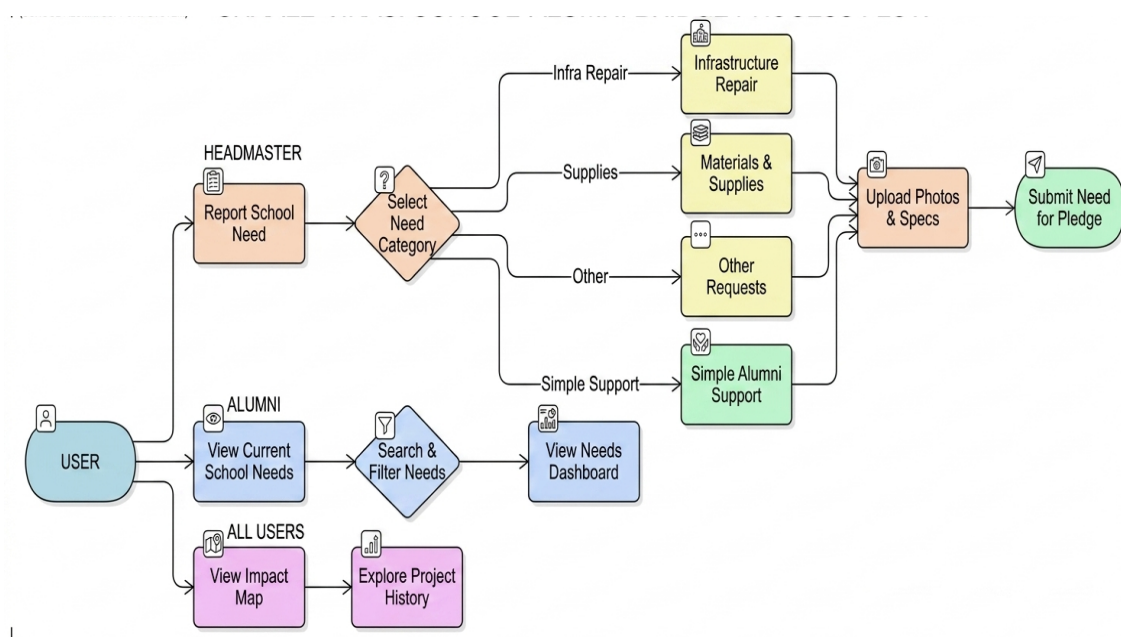


Fig 3.1 Navigation Design

Navigation Design shown in **Fig 3.1** represents the workflow of the **Shaale-Vikas** application, illustrating how users interact with different functionalities. The user can view school infrastructure needs, check details such as cost and progress, and contribute through a simulated pledge system. Users can also add new needs (for admin functionality in the prototype) and track the funding progress of each requirement. Additionally, the system provides a structured dashboard to help users easily navigate between available needs and understand their status, ensuring transparency and clarity in the contribution process.

3.4.3 Data Structuring and State Management

The application uses structured data models such as the **Need model class** to manage core data including need title, cost, collected amount, and progress details. State management is handled using Android activity lifecycle methods and UI update functions, ensuring that the interface dynamically reflects changes when new needs are added or when pledges are made. Local data handling is used to temporarily store and manage sample needs and contribution updates during the prototype stage. Data is maintained using ArrayLists and displayed efficiently using ListView/RecyclerView components with adapter-based UI handling. This approach ensures proper data organization, smooth UI updates, consistency across screens, and an efficient user experience throughout the application. The modular structure also supports future integration with Firebase for real-time data synchronization.

3.4.4 Generative AI-Assisted Development Workflow

An important aspect of the development process was the structured workflow followed to design and implement the Shaale-Vikas application. The development was carried out in an iterative manner, starting with planning and requirement analysis, followed by implementation of individual modules such as the needs dashboard, pledge system, and progress tracking features. Each feature was developed step-by-step to ensure proper integration and functionality. The system was continuously tested during development to maintain consistency across modules and improve usability. This structured approach helped in building a stable prototype while allowing scope for future enhancements such as Firebase integration and real-time updates.

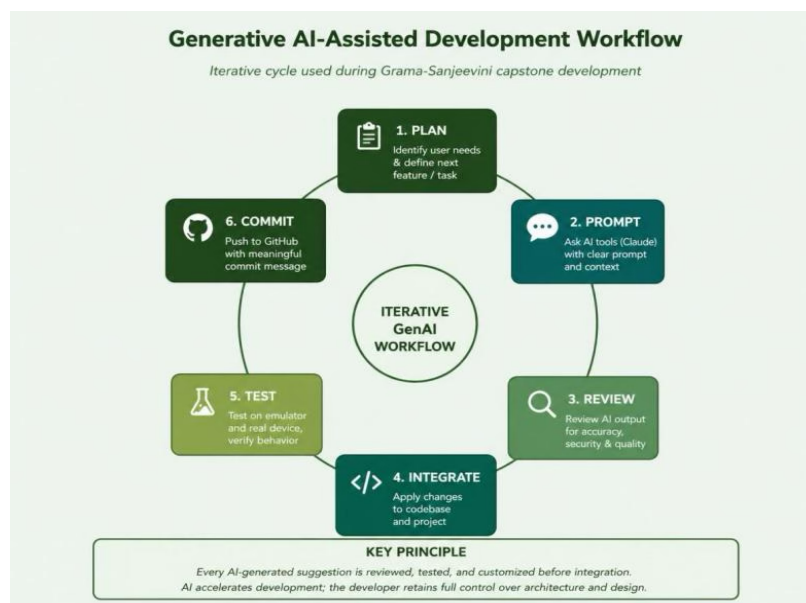


Fig 3.2 Generative AI-Assisted Development Workflow

In the Fig 3.2 represents the development cycle followed during the project, including planning, generating solutions, reviewing outputs, implementing features, testing, and final integration.

3.5 Implementation

3.5.1 Objectives

- To develop an Android application for managing school infrastructure needs.
- To enable users (alumni) to view and support school requirements through a digital platform.
- To allow school authorities to add and manage micro-level infrastructure needs.
- To implement a pledge-based system for simulated contributions.
- To display real-time progress of each need using visual indicators.
- To provide a user-friendly interface for viewing and tracking school requirements.
- To promote transparency and community participation in school development.
- To support structured data handling for efficient need management.

3.5.2 System Architecture Implementation

The system architecture of the Shaale -Vikas application shown in **Fig 3.3** is designed using a layered approach consisting of the mobile application layer, data handling layer, and user interaction layer. The process begins with the user interacting with the mobile application to view school infrastructure needs, check progress details, or make simulated pledges toward specific requirements.

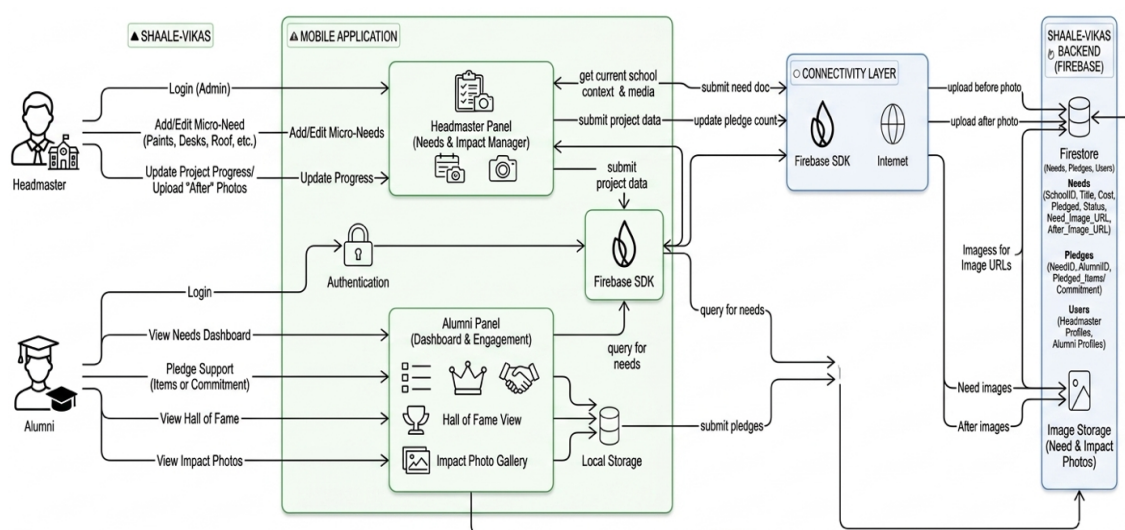


Fig 3.3 System Architecture Implementation

Within the mobile application layer, multiple components work together to process and manage user interactions efficiently. The dashboard module displays all listed school needs along with details such as title, estimated cost, collected amount, and progress status. The add-need module allows administrators to create and manage new infrastructure requirements, while the pledge module enables users to contribute through a simulated contribution system. Whenever a pledge is made, the application dynamically updates the collected amount and progress percentage in the interface.

The application also uses local data handling mechanisms to manage sample data and user interactions during the prototype stage. Data is organized using model classes, ArrayLists, and adapter-based UI components to ensure structured data management and smooth rendering of information on the dashboard.

The architecture is designed in a modular and scalable manner, allowing future integration with backend services such as Firebase Realtime Database and Firebase Storage for real-time synchronization, cloud storage, and image upload functionalities. Overall, the system architecture ensures efficient data flow, smooth communication between modules, responsive UI updates, and reliable application performance, while also providing flexibility for future enhancements and scalability.

3.5.3 Module-wise implementation

The Shaale – Vikas application is developed using a modular approach, where each functionality is implemented as a separate module to ensure better organization, scalability, and maintainability of the system.

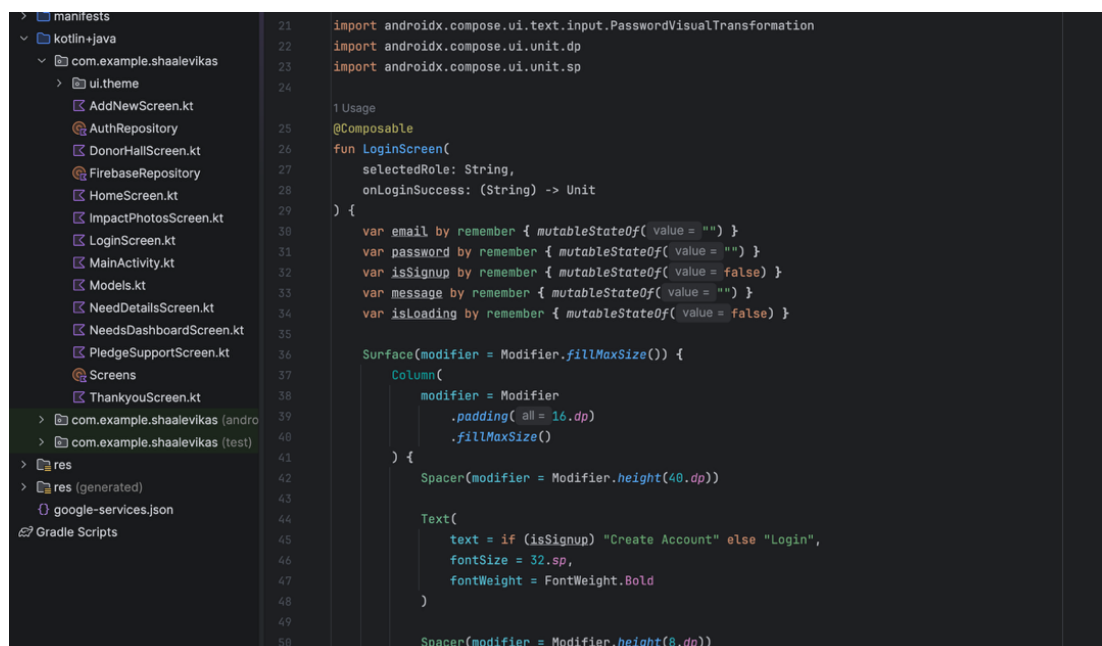


Fig 3.4 Login Module

The **Login Module** shown in **Fig 3.4** allows users to enter the application and select their role such as Alumni or Admin. This module manages basic user authentication flow and acts as the entry point of the application.

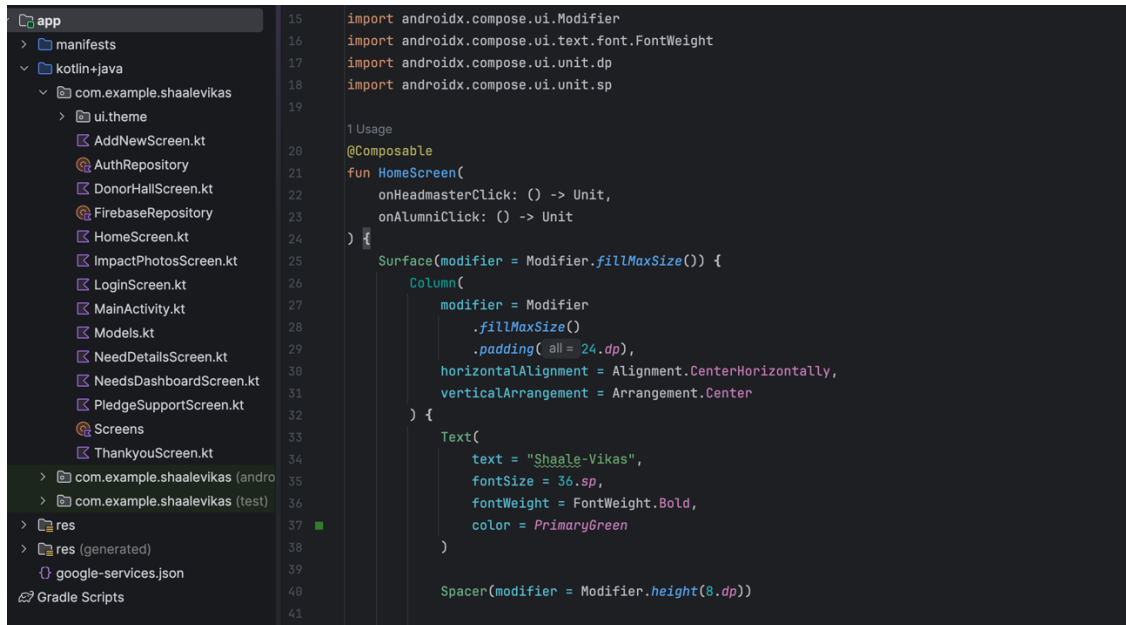


Fig 3.5 Home Screen Module

The **Home Screen Module** shown in **Fig 3.5** provides navigation to different sections of the application. It acts as the central interface from which users can access the dashboard, donor hall, impact photos, and other features.

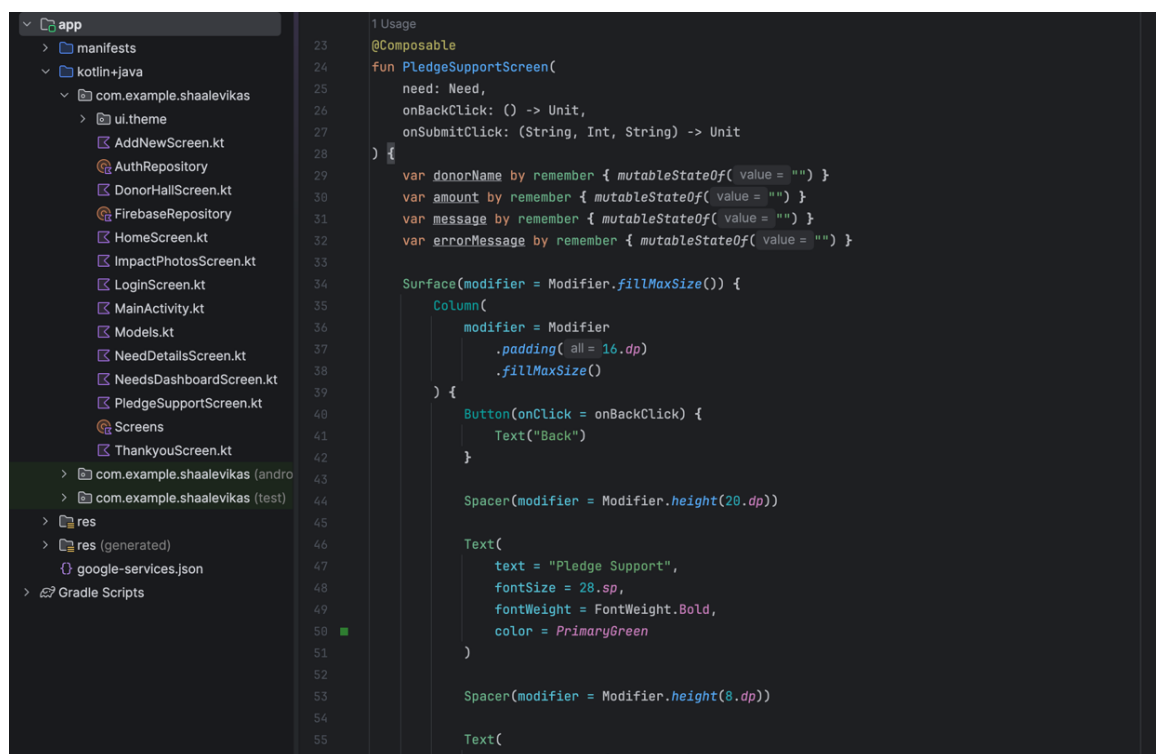


Fig 3.6 Pledge Support Module

The **Pledge Support Module** shown in **Fig 3.6** allows alumni users to contribute through a simulated pledge system. When a user makes a pledge, the collected amount and progress indicators are updated dynamically within the application.

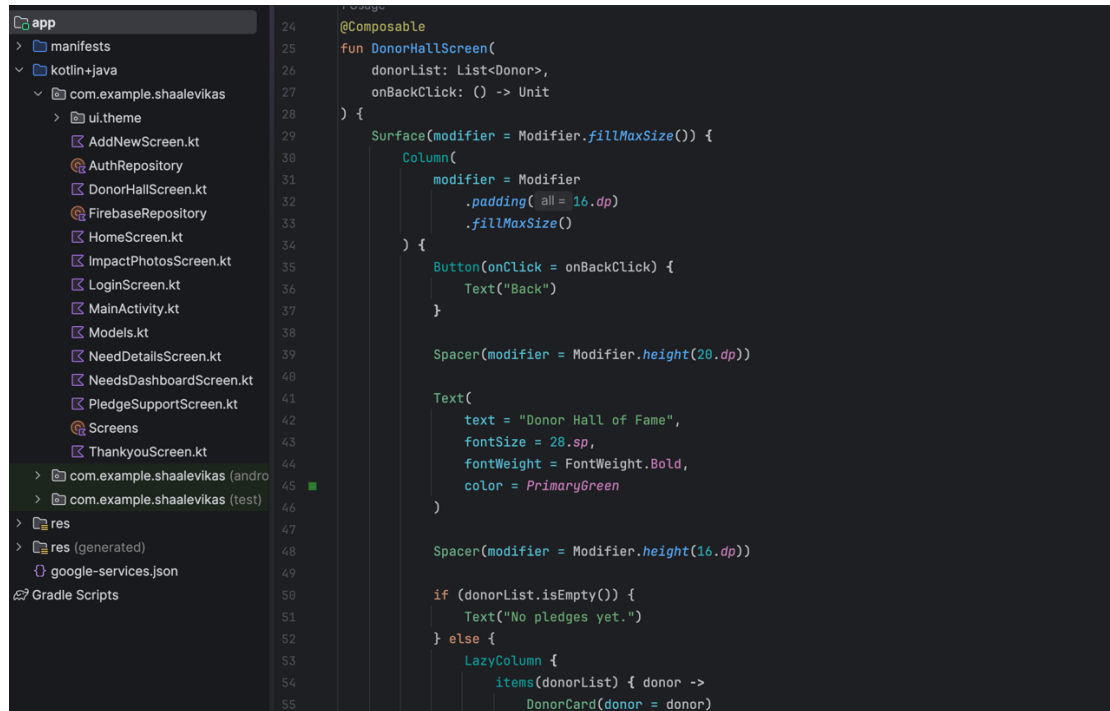


Fig 3.7 Donor Hall Module

The **Donor Hall Module** shown in **Fig 3.7** displays the list of contributors who have supported different school needs. This module helps promote transparency and encourages community participation.

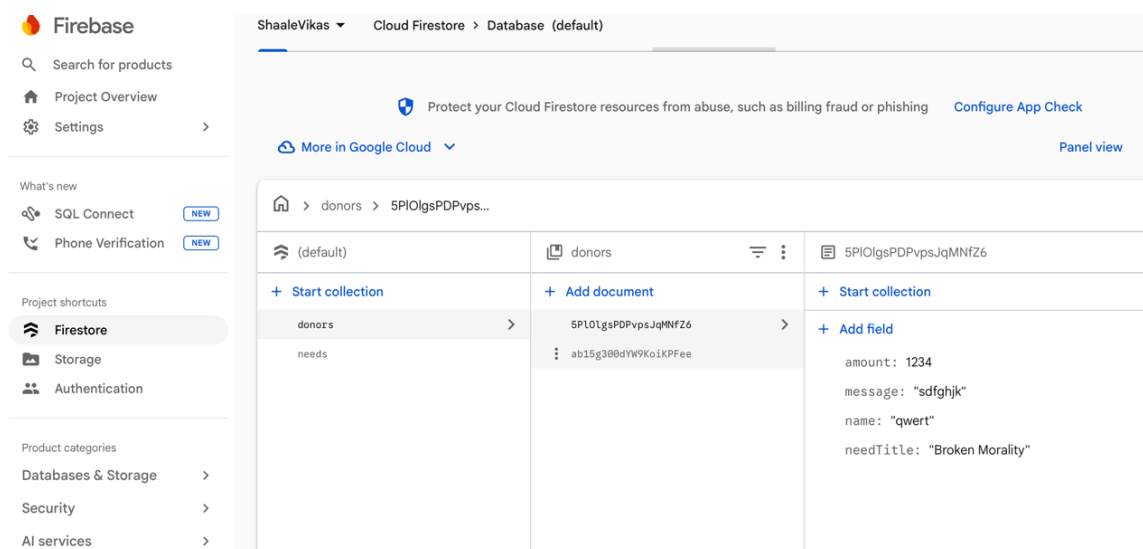


Fig 3.8 Firebase Repository Module

The **Firestore Repository Module** shown in **Fig 3.8** is structured to manage backend operations and future cloud integration. It is designed to support real-time data synchronization and centralized data management in future versions of the application.

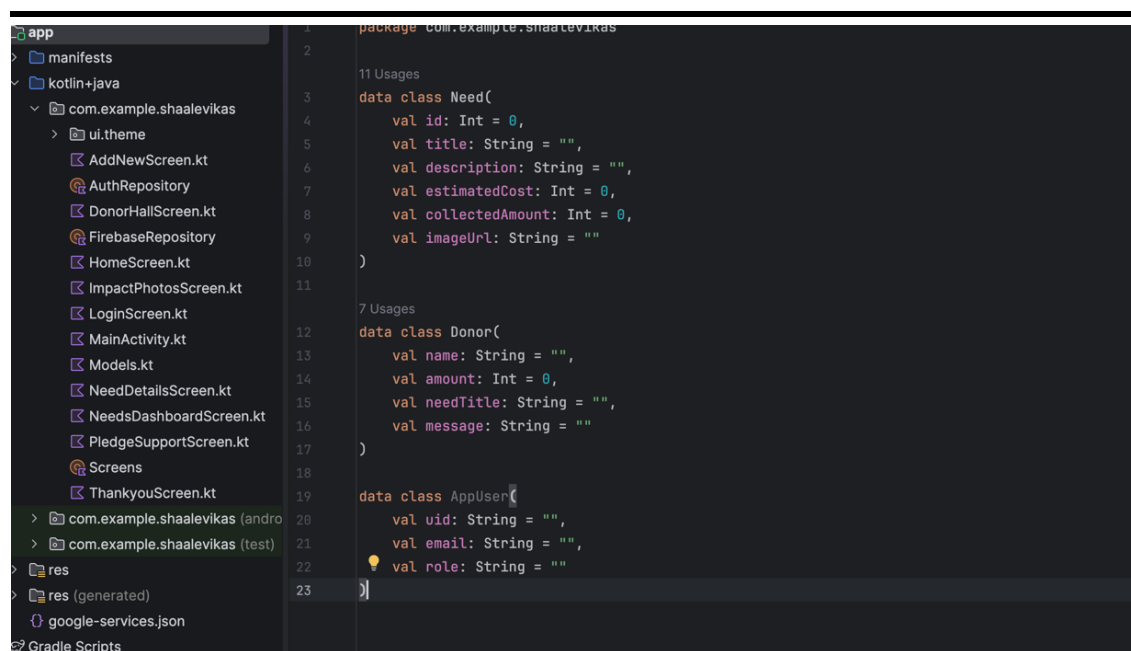


Fig 3.9 Models Module

The **Models Module** shown in **Fig 3.9** manages structured data classes used throughout the application. These models organize information related to needs, users, pledges, and other application data to ensure consistency and efficient data handling.

3.5.4 Transaction Management Logic

The application implements structured logic for managing school infrastructure needs and simulated contribution activities. When an administrator adds a new need, the system stores details such as need title, estimated cost, and collected amount within the application data structure. When a user makes a pledge, the system updates the collected amount and recalculates the progress percentage dynamically. The updated values are immediately reflected on the dashboard, ensuring consistency and real-time feedback within the application interface. The transaction flow is designed to maintain proper synchronization between user actions, displayed progress, and stored application data, ensuring smooth and reliable functionality throughout the system.

3.5.5 Analytics and Reporting

The application includes basic tracking and reporting features to help users understand the progress of school development activities. The dashboard displays details such as total needs, collected amount, and funding progress for each requirement. Users can view the status of different infrastructure needs and monitor contribution progress through visual indicators and structured lists. The Donor Hall module also helps in recognizing contributors and promoting transparency within the system. These reporting

features improve user engagement, provide better visibility of ongoing requirements, and encourage community participation in school development activities.

3.5.6 Error Handling and Validation

Error handling and validation are implemented to ensure the reliability and smooth functioning of the application. Input validation is performed to prevent invalid or incomplete data such as empty need titles or incorrect cost values. The system provides appropriate feedback to users through Toast messages and input checks whenever invalid actions are detected. Basic exception handling mechanisms are also implemented to prevent application crashes and ensure stable operation during user interactions such as adding needs or making pledges. These validations help maintain data consistency and improve overall usability of the application.

3.5.7 Testing and Debugging

The application was tested to ensure that all features function correctly under different scenarios. Testing was carried out using both Android Emulator and physical Android devices to verify application performance, responsiveness, and stability. Core functionalities such as displaying needs, updating progress values, button interactions, and navigation between screens were tested and refined. Minor UI issues and logical errors identified during development were corrected to ensure smooth operation and a better user experience. This testing process helped improve application reliability, interface consistency, and overall functionality of the Shaale-Vikas system.

3.6 Platform

3.6.1 Programming Language

The Shaale – Vikas application is developed using Kotlin, a modern and officially supported programming language for Android development. Kotlin is preferred due to its concise syntax, improved readability, and enhanced safety features compared to traditional languages like Java. It reduces boilerplate code and minimizes common programming errors through features such as null safety and type inference. Kotlin also provides seamless integration with Android libraries and frameworks, making it efficient for building scalable and maintainable applications. Its support for asynchronous programming further improves performance and responsiveness of the application. Overall, Kotlin plays a crucial role in ensuring clean code structure and efficient development of the application. Additionally, Kotlin enhances developer productivity by

simplifying application logic and improving overall code maintainability.

3.6.2 Library and Tools

The development of the Shaale-Vikas application utilizes various tools and libraries to enhance functionality and performance. Android Studio is used as the primary development environment, providing a comprehensive platform for coding, testing, and debugging. XML layouts are used for designing the user interface, ensuring a structured, responsive, and user-friendly experience. The application uses Android UI components such as ListView/RecyclerView, Buttons, and Progress indicators to display and manage school infrastructure needs efficiently. AndroidX libraries and Material Design components are also utilized to maintain a modern and consistent interface design. Local data handling and adapter-based UI management are used during the prototype stage to ensure smooth interaction and dynamic updates within the application. The overall architecture is designed to support future integration with Firebase services for real-time data synchronization and cloud storage. These tools and technologies collectively contribute to building a reliable, scalable, and efficient application.

CHAPTER 4

RESULTS

4.1 Project Results

Shaale-Vikas is the capstone project developed during this internship — an Android-based application designed to bridge the gap between rural schools and their alumni network. The application allows school authorities to post infrastructure needs such as repairs, furniture requirements, and other essential facilities, enabling alumni to view and support these needs through a transparent digital platform. The application addresses the common issue of delayed infrastructure support in rural schools by providing a simple and accessible system for showcasing current school requirements and tracking contribution progress. The system includes features such as a Needs Dashboard, pledge support system, progress tracking, donor recognition, and structured display of school requirements. In addition to contribution tracking, the application provides modules for viewing need details, managing school requirements, and displaying donor participation. The progress of each requirement is updated dynamically, helping users understand the current funding status clearly and transparently. The application is designed to be lightweight, user-friendly, and easy to navigate for both administrators and alumni users. By combining structured data handling with a responsive Android interface, Shaale-Vikas demonstrates the effective use of mobile application development to address real-world educational infrastructure challenges and promote community-driven school development.

4.2 Needs Dashboard Functionality

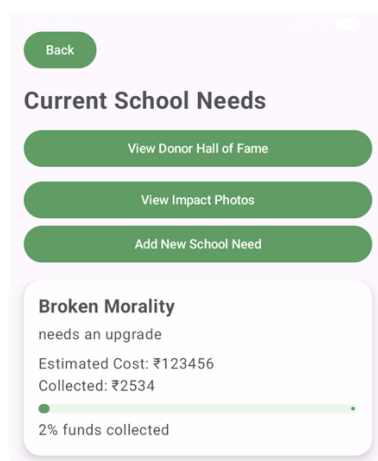


Fig 4.1 Needs Dashboard

The **Needs Dashboard** shown in **Fig 4.1** serves as the main interface of the Shaale-Vikas application, displaying all school infrastructure requirements in a structured and

user-friendly format. Each listed need contains details such as the title of the requirement, estimated cost, collected amount, and progress status. The developed application successfully enables users to view and manage school infrastructure needs through the Needs Dashboard. The dashboard efficiently displays requirements such as repairs, furniture needs, and other school-related necessities along with details including estimated cost, collected amount, and progress status. Users can interact with the system by selecting needs and making simulated pledges, after which the progress values are updated dynamically within the application. The dashboard provides a clear and structured interface that allows users to easily track ongoing requirements and understand the current funding progress. The implemented functionality ensures smooth user interaction, accurate display of data, and efficient handling of contribution updates, thereby improving transparency and community participation in school development activities.

4.3 Login Screen Functionality

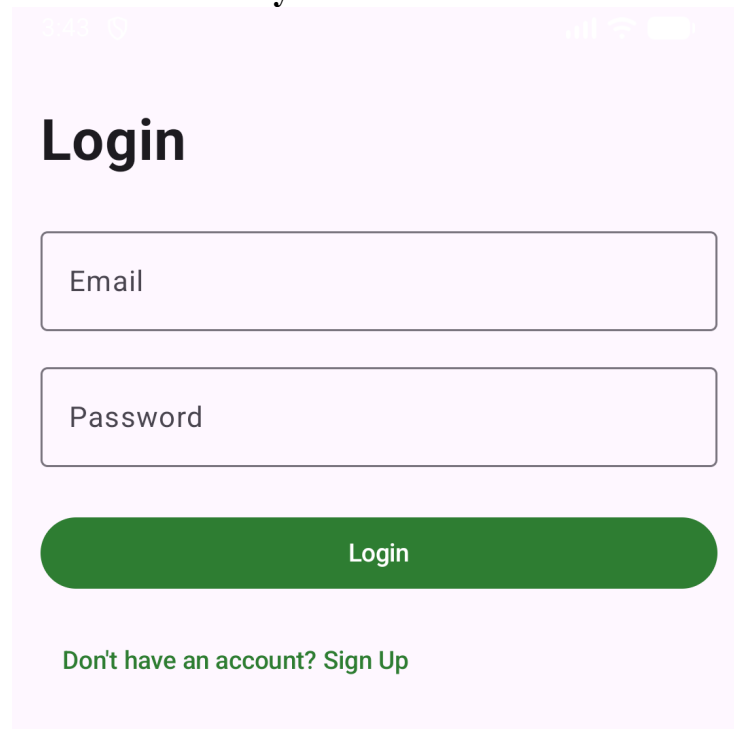
A mobile application login screen with a light purple background. At the top, the word "Login" is displayed in a large, bold, black font. Below it are two white rectangular input fields with rounded corners, each with a thin purple border. The first field is labeled "Email" and the second is labeled "Password". Below these fields is a large, rounded green button with the word "Login" in white text. At the bottom, there is a link that says "Don't have an account? Sign Up" in a smaller green font.

Fig 4.2 Login Screen

The Login Screen shown in **Fig 4.2** acts as the entry point of the Shaale-Vikas application. It allows users to access the system by selecting their role such as Admin or Alumni/User. The screen is designed with a simple and user-friendly interface to ensure easy navigation for all users. The module validates basic user input and redirects users to the appropriate dashboard based on the selected role. Proper UI components such as buttons, text fields, and role selection options are used to provide smooth interaction and improve accessibility. The login functionality ensures organized access flow within the

application and provides the foundation for role-based interaction across different modules of the system.

4.4 Home Screen Functionality

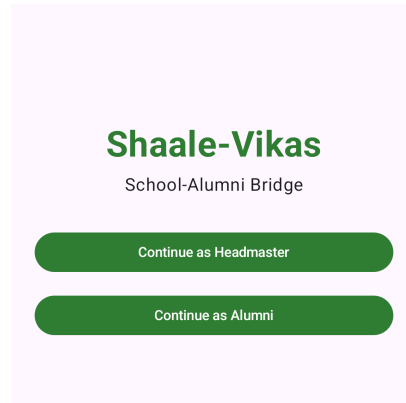


Fig 4.3 Home Screen

The Home Screen shown in **Fig 4.3** serves as the central navigation interface of the Shaale-Vikas application. It provides users with quick access to important modules such as the Needs Dashboard, Add Need section, Donor Hall, and Impact Photos. The screen is designed using a clean and structured layout to ensure smooth user interaction and easy accessibility. Interactive buttons and card-based components are used to improve navigation and enhance the overall user experience. The Home Screen helps users efficiently explore different functionalities of the application while maintaining a consistent and user-friendly interface design.

4.5 Donor Hall of Fame Functionality

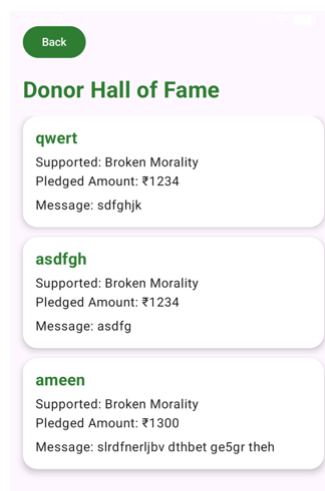


Fig 4. 4 Donor Hall of Fame

The Donor Hall of Fame module shown in **Fig 4.4** displays the list of contributors who have supported different school infrastructure needs through the pledge system. This feature is designed to recognize and appreciate alumni participation in school development

activities. The module presents donor information in a structured and visually organized format, helping users easily view contributor details and supported initiatives. It also promotes transparency and encourages greater community involvement by highlighting active participation within the platform. The implementation of this module enhances user engagement and strengthens the connection between schools and their alumni network.

4.6 Performance Evaluation

The Shaale-Vikas application demonstrates good performance in terms of responsiveness, usability, and stability. Core operations such as displaying school needs, updating pledge progress, and navigating between screens are executed smoothly without noticeable delays. The use of structured data handling and adapter-based UI components ensures efficient rendering of dashboard information and contribution updates. The application performs reliably under normal usage conditions and provides a smooth user experience across both emulator and physical Android devices. The lightweight design and organized architecture contribute to stable application performance during repeated interactions.

4.7 Advantages of the System

The Shaale-Vikas application offers several advantages by providing a simple and transparent platform for connecting schools with their alumni network. It allows school authorities to display current infrastructure needs while enabling alumni to support these requirements through a structured pledge system. The application improves transparency by displaying progress updates and contribution details clearly through visual indicators and dashboard components. Its user-friendly interface ensures easy navigation for users with different levels of technical knowledge. The system also promotes community participation, encourages alumni engagement in school development activities, and reduces communication gaps between schools and supporters.

4.8 Limitations

Despite its effectiveness, the application has certain limitations. The current version functions primarily as a prototype and uses local data handling instead of complete real-time cloud synchronization. Advanced features such as secure user authentication, real payment integration, and role-based access control are not fully implemented. Additionally, the system currently supports only basic progress tracking and does not include advanced analytics or notification features. These limitations may affect scalability

and broader deployment in large-scale environments. Furthermore, the application currently depends heavily on stable internet connectivity for smooth communication between users and database services. In areas with poor network coverage, performance and data synchronization may be affected. The user interface is designed mainly for demonstration purposes and may require further optimization to improve accessibility, responsiveness, and user experience across different Android devices. The system also lacks multilingual support, which could limit usability among users from diverse regional backgrounds. Data backup and recovery mechanisms are minimal in the current implementation, increasing the risk of data loss in unexpected situations. Moreover, security measures such as encryption, secure API handling, and advanced database protection are yet to be fully integrated. Future improvements are necessary to enhance reliability, security, scalability, and overall system efficiency for real-world deployment.

4.9 Future Enhancements

Future improvements to the application may include integration with Firebase for real-time database synchronization and cloud storage support. User authentication and role-based access control can be implemented to improve security and personalized access. Additional enhancements such as notification systems, real-time updates, image upload functionality, and advanced analytics dashboards can further improve user experience. The application can also be extended with digital payment integration and automated reporting features to support larger-scale school development initiatives.

4.10 Conclusion

In conclusion, the Shaale-Vikas application successfully achieves its objective of providing a simple and effective platform for connecting rural schools with their alumni network. The system demonstrates a user-friendly interface, structured data handling, and efficient implementation of features such as needs management, pledge tracking, and contribution visualization. The project highlights the effective use of Android application development in solving real-world educational infrastructure challenges through community participation and transparency. Overall, Shaale-Vikas serves as a practical and scalable solution that encourages collaborative school development and digital empowerment.

CHAPTER 5

COURSE OUTCOMES AND CO-PO MAPPING

5.1 COURSE OUTCOMES

Course title : Industry Internship(BCS803B)

CO	Description
CO1	Apply Python programming, Django framework, database management, and web development concepts to develop the College ERP System.
CO2	Demonstrate professional software development practices including project planning, documentation, debugging, testing, communication, and time management during project development.
CO3	Use modern web technologies and tools such as Django, SQLite, HTML, CSS, Bootstrap, JavaScript, and VS Code for application development.
CO4	Prepare technical documentation, implementation details, system architecture, and effectively present the developed College ERP System and project outcomes.
CO5	Prepare technical documentation, implementation details, system architecture, and effectively present the developed College ERP System and project outcomes.

5.2 PROGRAM OUTCOMES (POs)

PO	Description
PO1	Engineering Knowledge
PO2	Problem Analysis
PO3	Design/Development of Solutions
PO4	Conduct Investigations of Complex Problems
PO5	Engineering Tool Usage

PO6	The Engineer and The World
PO7	Ethics
PO8	Individual and Collaborative Team Work
PO9	Communication
PO10	Project Management and Finance
PO11	Life-Long Learning

5.3 CO – PO MAPPING

Mapping Levels

3 - High

2 – Medium

1- Low

Blank – No Significant Mapping

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	3	1	3	1	–	1	–	–	2
CO2	2	3	3	2	2	2	–	1	–	–	1
CO3	–	1	1	–	1	2	3	3	3	2	2
CO4	2	2	2	2	3	1	–	1	1	–	2
CO5	1	1	–	–	1	–	–	2	3	1	2

BIBLIOGRAPHY

- [1.] <https://developer.android.com/>
- [2.] <https://developer.android.com/compose>
- [3.] <https://kotlinlang.org/docs/home.html>
- [4.] <https://developer.android.com/develop/ui>
- [5.] <https://developer.android.com/develop/ui/views/layout/recyclerview>
- [6.] <https://firebase.google.com/docs/android/setup>
- [7.] <https://m3.material.io/>
- [8.] <https://developer.android.com/guide/navigation/>
- [9.] <https://codelabs.developers.google.com/>

Standards and Industry References

- [10.] MindMatrix 2026 — Problem Brief: Shaale-Vikas (Education/Infrastructure), Android App Development using GenAI Track.
- [11.] Ministry of Education, Government of India
<https://www.education.gov.in/>
- [12.] Digital India Initiative – Government of India
<https://www.digitalindia.gov.in/>
- [13.] Samagra Shiksha – Integrated Scheme for School Education
<https://samagra.education.gov.in/>
- [14.] National Education Policy (NEP) 2020
National Education Policy 2020
- [15.] UNICEF – Education and School Infrastructure Resources
<https://www.unicef.org/education>